

# **SIMULATING POWER GRID CYBER-ATTACKS: UTILIZING V-I CORRELATION TO DIFFERENTIATE FDI ATTACKS FROM PHYSICAL FAULTS**

*A report submitted in partial fulfillment of the requirements for the Award of Degree of*

**Bachelor of Technology**  
In  
**Electrical Engineering**

Submitted By:

**Ujjwal Keshri (Reg. No. 23110546)**

Under the Guidance of:

**Prof. A. K. Barisal, Professor, SES**



**SCHOOL OF ELECTRICAL SCIENCES**  
ODISHA UNIVERSITY OF TECHNOLOGY AND RESEARCH,  
BHUBANESWAR-751029

2026

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ODISHA UNIVERSITY OF TECHNOLOGY AND RESEARCH  
Ghatikia, Bhubaneswar-751029

**School of Electrical Sciences**

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**CERTIFICATE**

This is to certify that the project report entitled "Simulating Power Grid Cyber-Attacks: Utilizing V-I Correlation to Differentiate FDI Attacks from Physical Faults" submitted by Ujjwal Keshri (Reg. No. 23110546) of School of Electrical Sciences, Odisha University of Technology and Research, Bhubaneswar, fulfils the requirement of the regulation relating to the nature and standard of the work for the award of the degree of Bachelor of Technology in Electrical Engineering for academic year 2025-26.

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School of Electrical Sciences

**Supervisor**  
Prof. A. K. Barisal, Professor SES

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ODISHA UNIVERSITY OF TECHNOLOGY AND RESEARCH

Ghatikia, Bhubaneswar-751029

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**DECLARATION**

I do hereby declare that the project report entitled "Simulating Power Grid Cyber-Attacks: Utilizing V-I Correlation to Differentiate FDI Attacks from Physical Faults" is a bona-fide work of study carried out by me under the guidance of Prof. A. K. Barisal, Professor, School of Electrical Sciences, Odisha University of Technology and Research, Bhubaneswar. It has been prepared for the fulfillment of the requirements of the degree of Bachelor of Technology in Electrical Engineering. The work has not been submitted for any other purpose.

PLACE:

Ujjwal Keshri (23110546)

DATE:

## **ACKNOWLEDGEMENT**

I am sincerely indebted and grateful to my project supervisor Prof. A. K. Barisal, Professor, School of Electrical Sciences, Odisha University of Technology and Research, Bhubaneswar, for his invaluable guidance, constant inspiration, encouragement, and unwavering support throughout each stage of this endeavour. Without his perceptive comments and suggestions, the successful completion of this work would not have been possible.

I am also greatly thankful to the Head of the School of Electrical Sciences for providing all necessary facilities required for this research work. I would also like to thank all the faculty members and staff of the department for their cooperation and assistance.

I extend my heartfelt gratitude to my parents and friends for their constant encouragement, love, and moral support throughout this journey. Above all, I thank the Almighty for bestowing His blessings upon me.

Ujjwal Keshri (23110546)

# ABSTRACT

Modern smart grids rely heavily on Supervisory Control and Data Acquisition (SCADA) systems for real-time monitoring and control, making them highly vulnerable to sophisticated False Data Injection (FDI) cyber-attacks. In such attacks, malicious actors intelligently alter sensor measurements to deceive grid operators without triggering conventional bad data detection alarms. A critical operational blind spot arises when an anomaly appears: traditional protective relays struggle to rapidly distinguish whether a sudden voltage deviation results from a stealthy cyber-attack or a genuine physical short-circuit.

This project implements and validates two MATLAB functions — `detect_fdi()` and `classify_physical_fault()` — providing a complete, real-time, physics-based anomaly detection system for a 5-bus transmission network simulated in MATLAB/Simulink. The `detect_fdi()` function classifies the system state as Normal (0), Cyber-Attack (1), or Physical Fault (2) using calibrated per-unit V-I thresholds ( $V_{\text{mean}} = 0.6872$  pu,  $I_{\text{mean}} = 0.3882$  pu,  $Z_{\text{mean}} = 1.7702$  pu), a voltage-boundary cyber/physical separator at 0.6769 pu, five FDI attack detection modes, and a weighted event-scoring counter with blanking. The `classify_physical_fault()` function sub-classifies physical faults into Symmetrical (code 1), Unsymmetrical (code 2), Reverse Power Flow (code 3), or Intermittent (code 4) using rolling one-cycle RMS buffers and a power-direction integrator.

Simulation results confirm correct classification across all tested attack modes and fault types, with physical fault detection within 150  $\mu\text{s}$  (3 samples) and cyber detection within 100  $\mu\text{s}$  (2 samples) for magnitude-based attacks. Six diagnostic figures — including the Cyber Data & Attack Window Overlay — provide comprehensive visual validation.

Keywords: V-I Correlation, FDI Attack, SCADA, Physical Fault, Smart Grid Security, Threshold Detection, Reverse Power Flow, Intermittent Fault, MATLAB/Simulink, 5-Bus System.

## LIST OF ABBREVIATIONS

| Abbreviation | Full Form                                |
|--------------|------------------------------------------|
| FDI          | False Data Injection                     |
| SCADA        | Supervisory Control and Data Acquisition |
| IDS          | Intrusion Detection System               |
| V-I          | Voltage-Current Correlation              |
| BDD          | Bad Data Detection                       |
| RTU          | Remote Terminal Unit                     |
| RPF          | Reverse Power Flow                       |
| L-G          | Line-to-Ground (Asymmetric Fault)        |
| 3PH          | Three-Phase Short-Circuit Fault          |
| RMS          | Root Mean Square                         |
| pu           | Per Unit                                 |
| CPS          | Cyber-Physical System                    |
| PMU          | Phasor Measurement Unit                  |
| SLD          | Single-Line Diagram                      |
| Ts           | Simulation Sample Period (50 $\mu$ s)    |
| CB           | Circuit Breaker                          |
| RPF          | Reverse Power Flow                       |

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# CHAPTER 1

## INTRODUCTION

### 1.1 OVERVIEW

Modern electric power systems have evolved into sophisticated cyber-physical systems where real-time data communication is as critical as the physical infrastructure itself. Supervisory Control and Data Acquisition (SCADA) systems form the nervous system of smart grids, continuously monitoring bus voltages, line currents, and power flows to enable automatic control and operator decision-making. However, this deep integration of information technology with physical infrastructure has introduced a new class of threats: cyber-attacks targeting the digital measurement layer without touching physical equipment.

Among the most dangerous is the False Data Injection (FDI) attack, in which a malicious actor injects carefully crafted erroneous values into the SCADA measurement stream. Unlike crude denial-of-service attacks, a well-designed FDI attack manipulates sensor readings in a way that is mathematically consistent with power flow equations, evading traditional bad data detection (BDD) algorithms. The result is that operators receive a falsified picture of the network state, potentially triggering incorrect protective relay operations or failing to respond to a genuine crisis.

This project designs, implements, and validates two MATLAB functions — `detect_fdi()` and `classify_physical_fault()` — that together provide a complete, real-time, physics-based cyber-physical anomaly detection and classification system for a 5-bus transmission network modelled in MATLAB/Simulink. The system outputs a classified system state (Normal, Cyber-Attack, or Physical Fault) in real time at every 50  $\mu$ s simulation step, and further sub-classifies physical faults into four distinct categories.

### 1.2 LITERATURE REVIEW

The vulnerability of power system state estimation to FDI attacks was formally characterised by Liu et al. (2009), who demonstrated that an attacker with knowledge of the network topology can construct injection vectors invisible to the conventional largest normalised residual BDD method. This triggered extensive research into both stealthier attacks and more robust detection mechanisms. Historically, vulnerability analysis relied on DC power flow models, which neglect reactive power and assume flat voltage profiles. Recent research shows that DC-model-based attacks applied to an

AC grid produce data inconsistencies likely to trigger AC bad data detectors. Accurate vulnerability assessments must therefore use the full nonlinear AC model.

A complementary direction has emerged around physics-based detection. Anwar et al. (2015) proposed a data-driven approach distinguishing cyber-attacks from physical faults via statistical signatures. Centralised dynamic state estimation frameworks (IEEE Xplore, 2025) track system states and flag inconsistencies. Saber et al. (2024) addressed fault-masking attacks on differential protection, showing adversaries can engineer attacks to conceal a co-occurring physical fault. The present work addresses the same challenge through a computationally lightweight, locally deployable V-I correlation approach requiring no network model, no historical training data, and no iterative computation.

### 1.3 PROBLEM STATEMENT

Modern smart grids are highly vulnerable to FDI cyber-attacks in which malicious actors alter SCADA sensor measurements to deceive grid operators. A critical operational blind spot occurs: traditional protective relays cannot quickly determine whether a sudden voltage anomaly results from a stealthy cyber-attack or a genuine physical short-circuit.

Misclassifying a cyber-attack as a physical fault triggers unnecessary, widespread power outages. Conversely, misclassifying a real short-circuit as a digital attack leaves critical infrastructure exposed to catastrophic physical damage. Furthermore, even correctly detecting a physical fault is insufficient — the fault type (symmetrical, unsymmetrical, RPF, intermittent) determines the appropriate protective relay action and cannot be inferred from a binary flag alone.

### 1.4 RESEARCH OBJECTIVES

- Model a standard 5-bus transmission network in MATLAB/Simulink and derive normalised per-unit steady-state baselines:  $V_{\text{mean}} = 0.6872$  pu,  $I_{\text{mean}} = 0.3882$  pu,  $Z_{\text{mean}} = 1.7702$  pu.
- Implement `detect_fdi()`: a real-time MATLAB function classifying system state as Normal (0), Cyber-Attack (1), or Physical Fault (2) using a V-I boundary at 0.6769 pu, five FDI detection modes, a weighted event-scoring counter, frozen-sample replay detector, noise-variance tracker, and RPF integrator.
- Implement `classify_physical_fault()`: a sub-classifier outputting fault codes 0–4 using rolling one-cycle RMS buffers (400 samples at  $T_s = 50$   $\mu$ s), phase unbalance metric (threshold 15%),

power-direction integrator (threshold  $-0.015 \text{ pu}\cdot\text{s}$ ), and fault-transition counter (threshold 3 events in 200 ms).

- Simulate five FDI attack modes: Fake-Short, Fake-Sag, Fake-Swell, Replay Attack, and Noise Injection — all injected exclusively into the SCADA measurement channel.
- Simulate four physical fault types: Symmetrical (3-phase), Unsymmetrical (L-G), Reverse Power Flow (RPF), and Intermittent — as true physical events on the transmission lines.
- Validate both functions across all tested conditions and produce six diagnostic figures including the Cyber Data & Attack Window Overlay (Figure 6) for visual ground-truth validation.

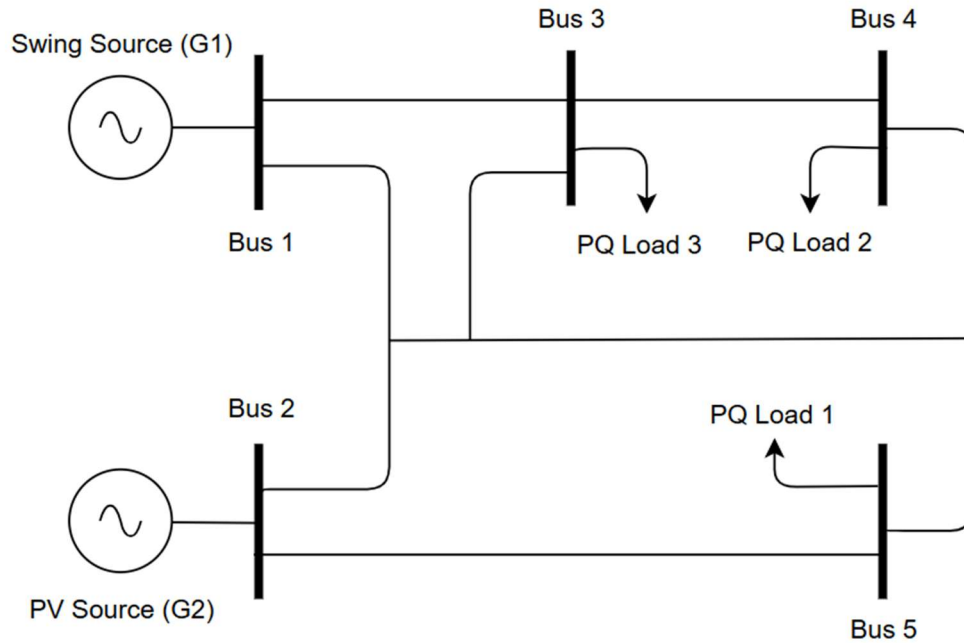
## CHAPTER 2

### SYSTEM ARCHITECTURE & DETECTION FUNDAMENTALS

#### 2.1 5-BUS TRANSMISSION NETWORK MODEL

The test system is a standard 5-bus transmission network, a widely used compact benchmark for power systems research. The network has five buses interconnected by six transmission lines, with Bus 1 (slack, 1.06 pu,  $0^\circ$ ) and Bus 2 (generator, 1.00 pu regulated) supplying three load buses — Buses 3, 4, and 5. Bus 5 is the primary monitoring bus where the FDI injection is applied.

All quantities are expressed in normalised per-unit (pu) on a 100 MVA base. The Simulink model uses the Specialized Power Systems toolbox with a fixed-step ode4 solver at  $T_s = 50 \mu s$ , providing 20,000 samples per second — sufficient to resolve fast electromagnetic transients during fault and attack events.



*Fig 2.1: Single-Line Diagram (SLD) of 5-Bus Transmission Network*

## 2.2 FDI ATTACK MECHANISM

An FDI attack targets the SCADA measurement layer. In the Simulink model, it is implemented as a Bias Injection block added between the physical voltage measurement output and the signal line feeding `detect_fdi()`. The attack operates exclusively on the signal path with zero connection to the physical power system. Therefore, true line currents remain physically unchanged — the central fact exploited by the V-I correlation detector.

Five distinct FDI attack modes are simulated: (1) Fake-Short: deflates apparent impedance via overcurrent-like signature on measured signal; (2) Fake-Sag: depresses measured voltage below  $V_{\text{sag\_th}}$  (0.6528 pu); (3) Fake-Swell: inflates measured voltage above  $V_{\text{swell\_th}}$  (0.7216 pu); (4) Replay Attack: freezes voltage samples for more than 5 consecutive steps (250  $\mu\text{s}$ ); (5) Noise Injection: superimposes high-variance signal noise on the current measurement path, detected by a rolling 20-sample variance buffer.

## 2.3 PHYSICAL FAULT CHARACTERISTICS

Physical faults cause a sudden, unintended change in the actual electrical network state. Four categories are simulated:

- Symmetrical (3-Phase) Fault: All three conductors connect simultaneously. Voltage collapses at the faulted bus, fault current surges, and phase current unbalance remains below 15%. `classify_physical_fault()` outputs Code 1.
- Unsymmetrical (L-G) Fault: One phase connects to ground, causing asymmetric voltage collapse and unequal fault currents across phases. Phase current unbalance exceeds 15%. `classify_physical_fault()` outputs Code 2.
- Reverse Power Flow (RPF) Fault: Caused by islanding conditions or generator loss events, where power flows in the reverse direction. The power-direction integrator `dir_integral` accumulates negative values and flags RPF when it drops below -0.015 pu·s. `classify_physical_fault()` outputs Code 3.
- Intermittent Fault: A fault that clears and re-strikes 3 or more times within 200 ms (4000 samples at  $T_s = 50 \mu\text{s}$ ). Tracked via a flag-transition counter. `classify_physical_fault()` outputs Code 4 — with the highest priority, overriding all other codes.

## 2.4 V-I CORRELATION DETECTION PRINCIPLE

The detection principle: a physical fault produces a correlated V-I response (voltage drops and current rises simultaneously), whereas an FDI attack produces an uncorrelated response (voltage signal changes, physical current unchanged).

The critical separator in `detect_fdi()` is the voltage boundary at 98.5% of  $V\_mean\_base$ :

$$is\_phys\_voltage = (V_{rms} < 0.985 \times 0.6872) = (V_{rms} < 0.6769 \text{ pu}) \quad [\text{Physical domain}]$$

$$is\_cyber\_voltage = (V_{rms} \geq 0.985 \times 0.6872) = (V_{rms} \geq 0.6769 \text{ pu}) \quad [\text{Cyber domain}]$$

Any impedance or current anomaly observed while `is_cyber_voltage` is true is treated as an FDI event. The same anomaly observed while `is_phys_voltage` is true is treated as a physical fault. This elegant single-boundary rule is augmented by the weighted cyber-event scoring counter and the RPF integrator for comprehensive robustness against all five attack types.

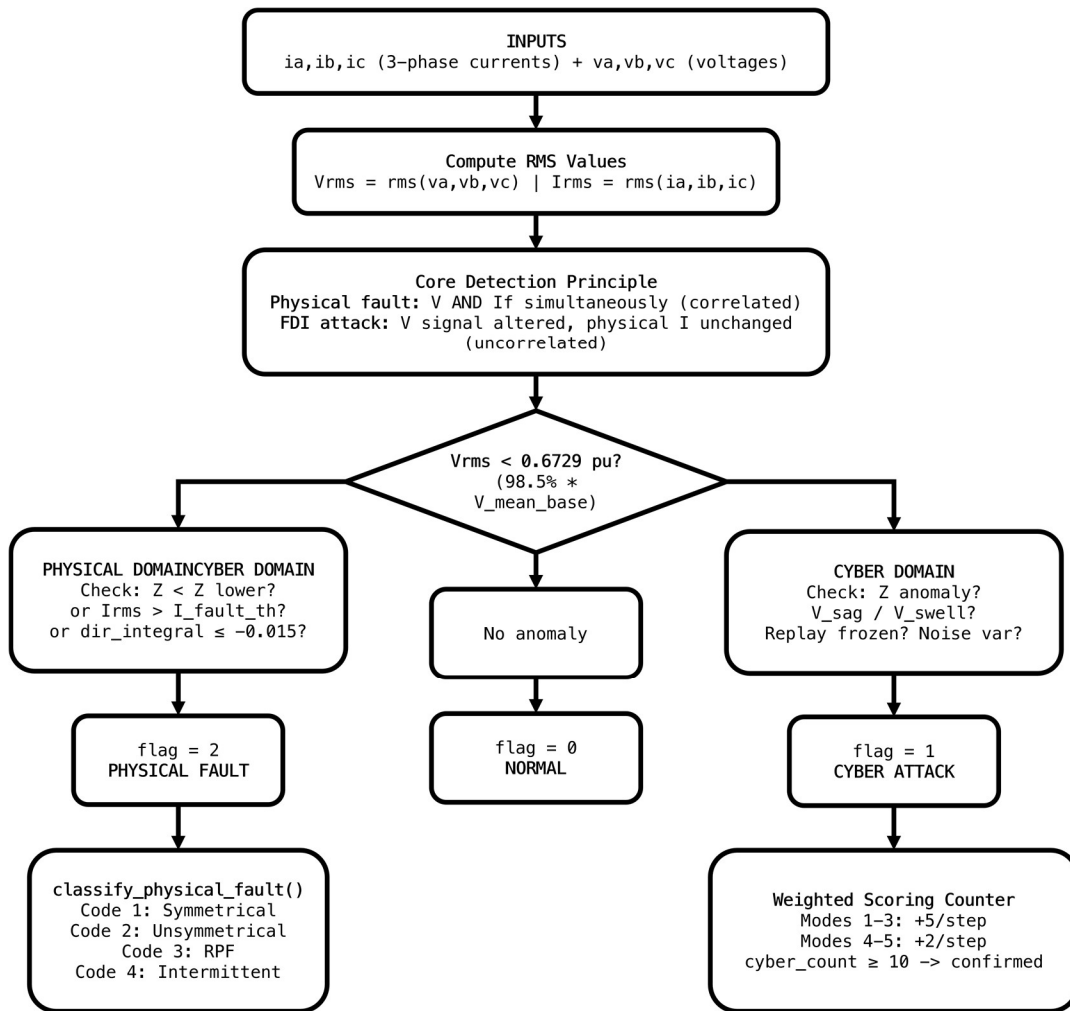


Fig 2.2: V-I Correlation Detection Logic — Cyber-vs-Physical Boundary Conceptual Diagram

## CHAPTER 3

### ALGORITHM DESIGN & THRESHOLD DERIVATION

#### 3.1 SYSTEM SPECIFICATIONS

| Parameter                     | Value                                       |
|-------------------------------|---------------------------------------------|
| System Base MVA               | 100 MVA                                     |
| Number of Buses               | 5 (2 Generator + 3 Load)                    |
| Number of Transmission Lines  | 6                                           |
| Bus 1 Voltage (Slack)         | 1.06 pu, 0° reference                       |
| Bus 2 Voltage (Generator)     | 1.00 pu (regulated)                         |
| Total Active Load             | ~283 MW                                     |
| Simulation Sample Period (Ts) | 50 $\mu$ s $\rightarrow$ 20 kHz sample rate |
| Solver                        | ode4 Fixed-Step Runge-Kutta                 |
| Nominal Frequency             | 50 Hz                                       |
| RMS Buffer Length (classify)  | 400 samples = 20 ms = 1 full cycle at 50 Hz |

*Table 3.1: 5-Bus System Specifications*

#### 3.2 BASELINE PARAMETER DERIVATION

The three per-unit baseline constants embedded in `detect_fdi()` are derived from steady-state simulation of the 5-bus network. All thresholds are computed as fixed percentage margins on these baselines, ensuring the detector is self-calibrating relative to the specific network operating point.

| Baseline Parameter      | Symbol                   | Value (pu) | Physical Meaning                                                    |
|-------------------------|--------------------------|------------|---------------------------------------------------------------------|
| Mean Apparent Impedance | <code>Z_mean_base</code> | 1.7702 pu  | Steady-state V/I ratio at monitoring bus under normal load          |
| Mean Line Current       | <code>I_mean_base</code> | 0.3882 pu  | Normal operating RMS current magnitude                              |
| Mean Bus Voltage        | <code>V_mean_base</code> | 0.6872 pu  | Normal RMS voltage magnitude at Bus 5                               |
| Cyber/Physical Boundary | <code>V_boundary</code>  | 0.6769 pu  | 98.5% of <code>V_mean</code> — separates cyber from physical domain |

*Table 3.2: Normalised Per-Unit Baseline Parameters (Derived from `detect_fdi` Code Constants)*

### 3.3 IMPEDANCE & CURRENT THRESHOLD DESIGN

| Threshold  | Formula                                     | Computed Value                          | Purpose in detect_fdi()                                |
|------------|---------------------------------------------|-----------------------------------------|--------------------------------------------------------|
| Z_upper    | $1.25 \times Z_{\text{mean\_base}}$         | 2.2128 pu                               | High-impedance fault boundary                          |
| Z_lower    | $0.75 \times Z_{\text{mean\_base}}$         | 1.3277 pu                               | Low-impedance short-circuit boundary                   |
| I_fault_th | $1.30 \times I_{\text{mean\_base}}$         | 0.5047 pu                               | Overcurrent threshold for fault / fake-short detection |
| I_breaker  | $0.30 \times I_{\text{mean\_base}}$         | 0.1165 pu                               | Dead-bus / breaker-open suppression level              |
| V_sag_th   | $0.95 \times V_{\text{mean\_base}}$         | 0.6528 pu                               | Fake voltage sag detection lower boundary              |
| V_swell_th | $1.05 \times V_{\text{mean\_base}}$         | 0.7216 pu                               | Fake voltage swell detection upper boundary            |
| V_boundary | $0.985 \times V_{\text{mean\_base}}$        | 0.6769 pu                               | Cyber-vs-physical domain separator                     |
| Noise Thr  | $20 \times (0.02 \times I_{\text{mean}})^2$ | 0.001206                                | Noise injection rolling variance threshold             |
| RPF Thr    | dir_integral                                | $\leq -0.015 \text{ pu} \cdot \text{s}$ | Reverse power-flow accumulation gate                   |

Table 3.3: V-I Impedance Threshold Derivation (All Values Computed from detect\_fdi Constants)

### 3.4 PHYSICAL FAULT DETECTION CONDITIONS — detect\_fdi()

| Fault Type            | Z Condition             | I Condition                          | Voltage Gate                    | Confirmed When                                       |
|-----------------------|-------------------------|--------------------------------------|---------------------------------|------------------------------------------------------|
| Short-Circuit (Low-Z) | $Z < 1.3277 \text{ pu}$ | $I_{\text{rms}} > 0.5047 \text{ pu}$ | is_phys_voltage (Vrms < 0.6769) | fault_count $\geq 3$ (150 $\mu\text{s}$ )            |
| Reverse Power Flow    | —                       | —                                    | Any                             | dir_integral $\leq -0.015 \text{ pu} \cdot \text{s}$ |

Table 3.4: Physical Fault Detection Conditions in detect\_fdi()

### 3.5 CYBER ATTACK DETECTION MODES — detect\_fdi()

| Sl no. | Attack Mode     | Primary Condition                                                                      | Voltage Gate     | Score / Step |
|--------|-----------------|----------------------------------------------------------------------------------------|------------------|--------------|
| 1      | Fake-Short      | $Z < 1.3277 \text{ pu}$ AND $I_{\text{rms}} > 0.5047 \text{ pu}$                       | is_cyber_voltage | +5           |
| 2      | Fake-Sag        | $V_{\text{rms}} < 0.6528 \text{ pu}$ AND $I_{\text{rms}} > 0.5 \times I_{\text{mean}}$ | is_cyber_voltage | +5           |
| 3      | Fake-Swell      | $V_{\text{rms}} > 0.7216 \text{ pu}$ AND $I_{\text{rms}} > 0.2717 \text{ pu}$          | is_cyber_voltage | +5           |
| 4      | Replay Attack   | $\Delta v < 1e-3$ for > 5 consecutive steps                                            | — (independent)  | +2           |
| 5      | Noise Injection | mean(noise_var_buf) > 0.001206                                                         | — (independent)  | +2           |

Table 3.5: Cyber Attack Detection Modes & Conditions — detect\_fdi()

### 3.6 CYBER-EVENT WEIGHTED SCORING SYSTEM

| Event                                                 | Score Change           | Detection Threshold            | Blanking / Suppression                              |
|-------------------------------------------------------|------------------------|--------------------------------|-----------------------------------------------------|
| Modes 1–4 active (fake_short / fake_sag / fake_swell) | + 5 per step           | cyber_count $\geq$ 10          | Suppressed during fault_detected or breaker_open    |
| Modes 5–6 active (replay attack / noise injection)    | + 2 per step           | (same threshold)               | Suppressed during fault_detected or breaker_open    |
| No cyber condition active (natural decay)             | – 5 per step           | —                              | Applied always (no suppression)                     |
| Physical fault detected (blanking activation)         | cyber_count reset to 0 | Noise & frozen buffers flushed | 200-step (10 ms) holdoff applied after fault clears |
| Breaker open — dead bus ( $I_{rms} < 0.1165$ pu)      | cyber_count reset to 0 | Noise & frozen buffers flushed | Active while $I_{rms}$ below breaker threshold      |

Table 3.6: Cyber-Event Weighted Scoring System — *detect\_fdi()*

### 3.7 TIMING & COUNTER PARAMETERS

| Parameter                      | Code Constant              | Physical Duration | Effect                                                       |
|--------------------------------|----------------------------|-------------------|--------------------------------------------------------------|
| Sample Period (Ts)             | 50e-6 s                    | 50 $\mu$ s        | Simulation time step — all counts referenced to this         |
| Physical fault confirm         | fault_count $\geq$ 3       | 150 $\mu$ s       | Debounces single-sample transients from contact bounce       |
| Fault latch duration           | 250 steps                  | 12.5 ms           | Holds flag=2 after fault clears to suppress ring-on-recovery |
| Cyber holdoff post-fault       | 200 steps                  | 10.0 ms           | Blanks cyber accumulation after physical event               |
| Replay frozen threshold        | frozen_count $>$ 5         | 250 $\mu$ s       | Detects frozen / static voltage samples (replay attack)      |
| Noise buffer size              | 20 samples                 | 1.0 ms            | Rolling variance window for noise attack detection           |
| RPF integral threshold         | dir_integral $\leq$ -0.015 | N/A               | Accumulated reverse-power-flow trigger gate                  |
| Intermittent window (classify) | 4000 steps                 | 200 ms            | Window for counting fault re-strike transitions              |
| Intermittent trip count        | fluct_count $\geq$ 3       | N/A               | Minimum fault re-strikes to declare intermittent fault       |
| RMS rolling buffer (classify)  | 400 samples                | 20 ms (1 cycle)   | Circular RMS buffer for per-phase current magnitude          |

Table 3.7: All Timing & Counter Parameters (Derived from Both MATLAB Functions)

### 3.8 PHYSICAL FAULT SUB-CLASSIFICATION — `classify_physical_fault()`

The `classify_physical_fault()` function receives the flag output from `detect_fdi()` and is activated only when `flag = 2`. It outputs a `fault_code` (0–4) using a strict priority ladder:

| Code | Fault Type               | Decision Criterion                                                             | Priority                |
|------|--------------------------|--------------------------------------------------------------------------------|-------------------------|
| 0    | Normal — No Active Fault | <code>flag ≠ 2</code> (gate closed — function returns immediately)             | Gatekeeper              |
| 4    | Intermittent Fault       | <code>intermittent_latch = 1 → ≥ 3 flag=2 transitions within 200 ms</code>     | Highest (overrides 1–3) |
| 3    | Reverse Power Flow       | <code>rpf_event_latch = true → dir_integral ≤ -0.015 pu·s during flag=2</code> | Second (overrides 1–2)  |
| 2    | Unsymmetrical (L-G)      | Phase RMS unbalance > 15% of <code>I_avg_rms</code>                            | Third                   |
| 1    | Symmetrical (3-Phase)    | Phase RMS unbalance ≤ 15% of <code>I_avg_rms</code> (default physical)         | Lowest                  |

*Table 3.8: Physical Fault Sub-Classification Codes & Priority Ladder — `classify_physical_fault()`*

The phase current unbalance metric (Equation 8) determines the Code 1 vs Code 2 decision:

$$\text{unbalance} = \max(|I_{a\_rms} - I_{avg}|, |I_{b\_rms} - I_{avg}|, |I_{c\_rms} - I_{avg}|) / I_{avg\_rms} \dots(8)$$

where `Ia_rms`, `Ib_rms`, `Ic_rms` are computed from rolling 400-sample circular buffers (one full 50 Hz cycle). A threshold of 0.15 (15%) distinguishes balanced three-phase faults from asymmetric events.

## CHAPTER 4

### MATLAB/SIMULINK SIMULATION

#### 4.1 POWER SYSTEM MODEL IMPLEMENTATION

The 5-bus power system is built in MATLAB/Simulink using the Specialized Power Systems toolbox. Three-Phase V-I Measurement blocks at each bus simultaneously output instantaneous phase voltages and line currents. Three-Phase PI Section Line blocks model transmission lines with standard 5-bus test system parameters. The ode4 fixed-step solver runs at  $T_s = 50 \mu s$  throughout all simulations.

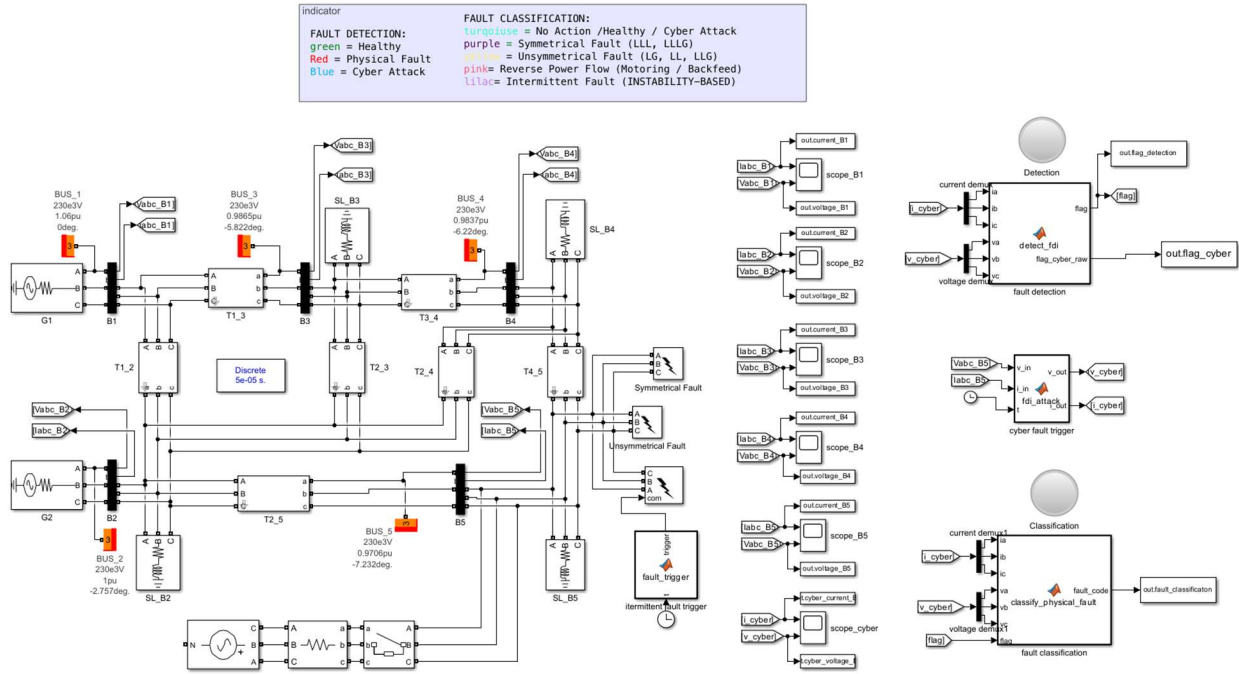


Fig 4.1: MATLAB/Simulink Model of 5-Bus System with *detect\_fdi()* and *classify\_physical\_fault()* MATLAB Function Blocks

#### 4.2 detect\_fdi() — SIMULINK INTEGRATION

The *detect\_fdi()* MATLAB Function block receives six scalar inputs: *ia*, *ib*, *ic* (instantaneous three-phase currents) and *va*, *vb*, *vc* (instantaneous three-phase voltages) from the Bus 5 measurement block. It outputs *flag* (0/1/2) and *flag\_cyber\_raw*. Persistent variables maintain all state across steps — *cyber\_count*, *fault\_count*, *fault\_latch*, *dir\_integral*, *frozen\_count*, *noise\_var\_buf*, and *is\_rpf\_active* — enabling the time-accumulation logic without external memory blocks.

### **4.3 classify\_physical\_fault() — SIMULINK INTEGRATION**

The `classify_physical_fault()` MATLAB Function block receives the same six V-I inputs plus the flag output from `detect_fdi()`. This cascaded architecture ensures sub-classification activates only when `flag = 2`. The rolling RMS buffers (`ia_buf`, `ib_buf`, `ic_buf` — each 400 elements) are implemented as circular arrays with `rms_buf_idx` tracking the write position. At each step, the oldest sample is overwritten and `mean()` computes the one-cycle squared average.

### **4.4 ATTACK & FAULT INJECTION BLOCKS**

FDI attacks: A Switch block selects between the real voltage measurement and a biased/modified version based on an attack-enable signal. Six attack configurations are implemented for Modes 1–6 as described in Table 3.5. All injection operates on signal paths only.

Physical faults: Three-Phase Fault blocks are placed on selected lines, configured for L-G (single-phase-to-ground) and 3-phase (bolted) events at  $t = 0.5$  s with 0.1 s duration. Intermittent faults are simulated by toggling the fault block ON/OFF three times within a 200 ms window.

## CHAPTER 5

### RESULT ANALYSIS

#### 5.1 NORMAL OPERATION RESULTS

The simulation runs for 1.0 s under normal load conditions. Bus voltages remain near rated values and Bus 5 Vrms stays near  $V_{\text{mean\_base}} = 0.6872$  pu. The impedance  $Z$  remains between  $Z_{\text{lower}}$  (1.3277 pu) and  $Z_{\text{upper}}$  (2.2128 pu). Both `detect_fdi()` outputs hold constant at 0 (Normal). `Classify_physical_fault()` outputs `fault_code` = 0 throughout.

| Bus No.               | Voltage Mag (pu) | Voltage Angle | Active Load (MW) | Reactive Load (MVAR) |
|-----------------------|------------------|---------------|------------------|----------------------|
| Bus 1 (Slack)         | 1.060            | 0.00°         | —                | —                    |
| Bus 2 (Gen)           | 1.000            | −4.98°        | —                | —                    |
| Bus 3 (Load)          | 0.987            | −12.72°       | 94.2             | 19.0                 |
| Bus 4 (Load)          | 0.984            | −10.33°       | 47.8             | −3.9                 |
| Bus 5 (Load, Monitor) | 0.972            | −8.78°        | 7.6              | 1.6                  |

Table 5.1: Normal Operating State — V-I Values at All Buses

##### 5.1.1 System Logic & Detection Flags — Figure 1 (Three-Panel)

Figure 1 is a three-panel diagnostic figure with a shared time axis. It must be placed as a composite figure with subplots stacked vertically (Fig 5.1A/B/C). These three signals together constitute the primary validation output of the detection system.

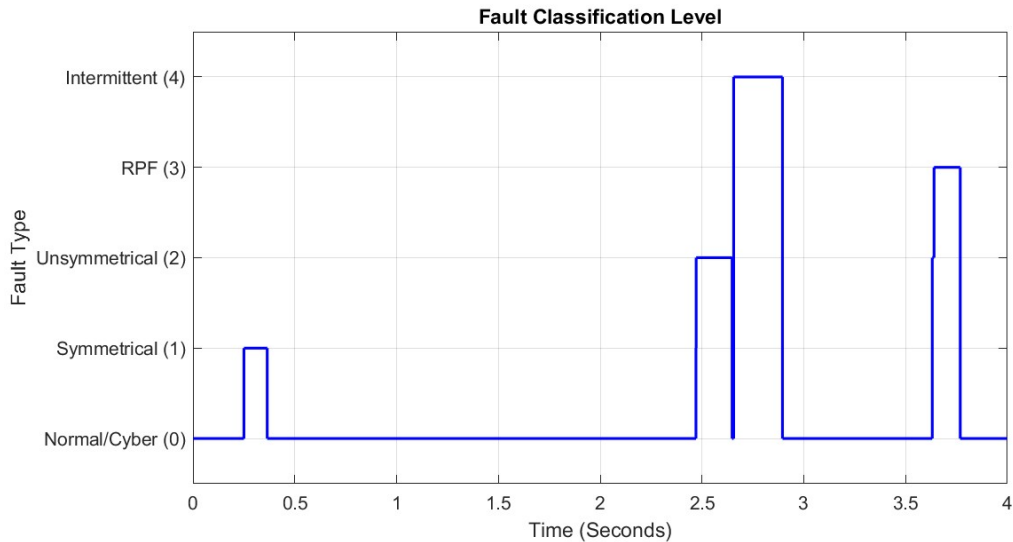
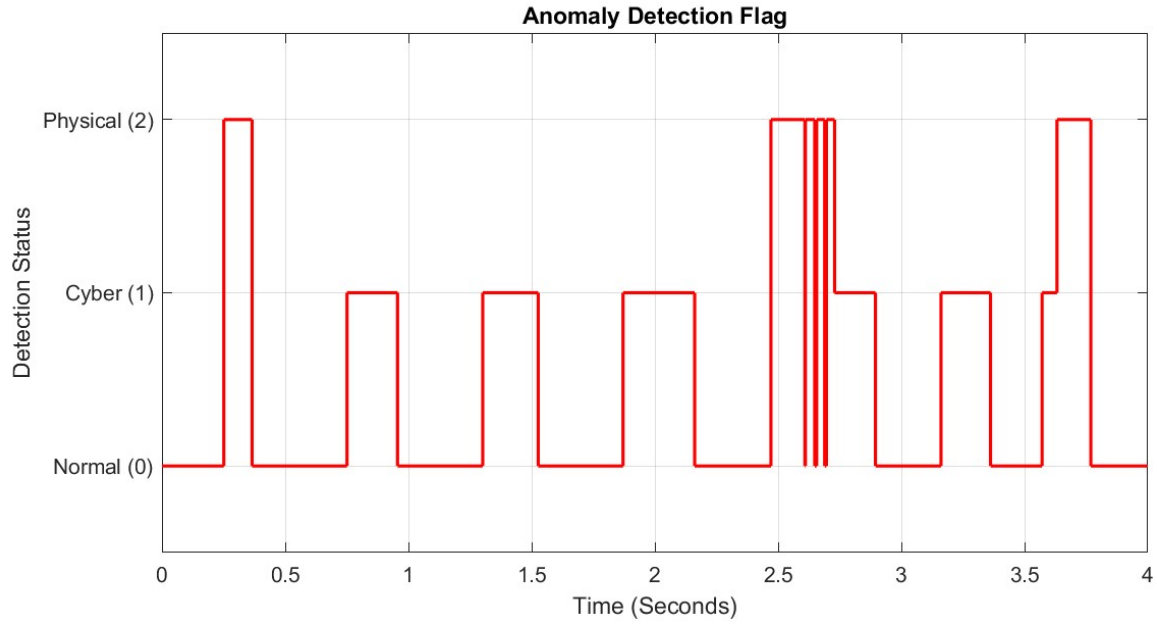
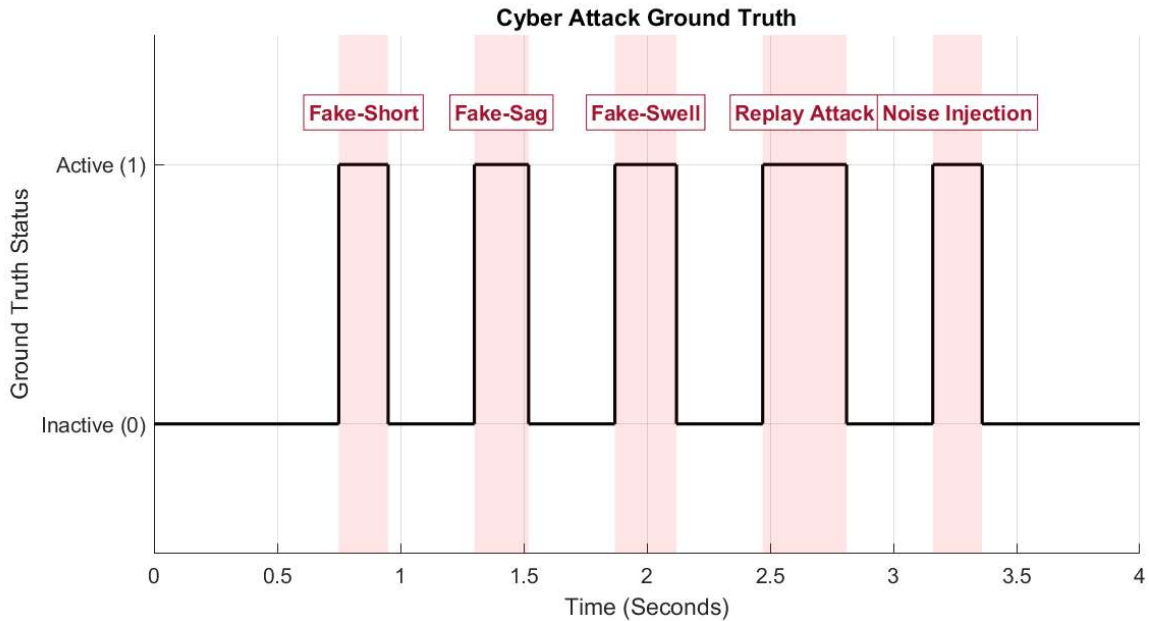


Fig 5.1A: Fault Classification Level (fault\_code: 0–4)



*Fig 5.1B: Anomaly Detection Flag (0=Normal, 1=Cyber, 2=Physical)*



*Fig 5.1C: Cyber Attack Ground Truth Window*

Plot 1A (fault\_code) rises to codes 1–4 during physical fault windows and returns to 0 otherwise. Plot 1B (flag) transitions to 1 during cyber-attack windows and to 2 during physical fault windows. Plot 1C (ground truth) shows the exact injection control signal — enabling pixel-perfect validation that detection flags align with actual attack windows. The latency between Plot 1C rising and Plot 1B rising to 1 directly measures the cyber detection delay.

## 5.2 FDI ATTACK RESULTS — ALL SIX MODES

FDI attacks are applied sequentially to Bus 5 voltage measurements. In all five modes, physical line currents remain unchanged. `detect_fdi()` raises flag to 1 within the scoring accumulation time. `classify_physical_fault()` outputs `fault_code = 0` throughout all attack windows (correct: no physical fault has occurred).

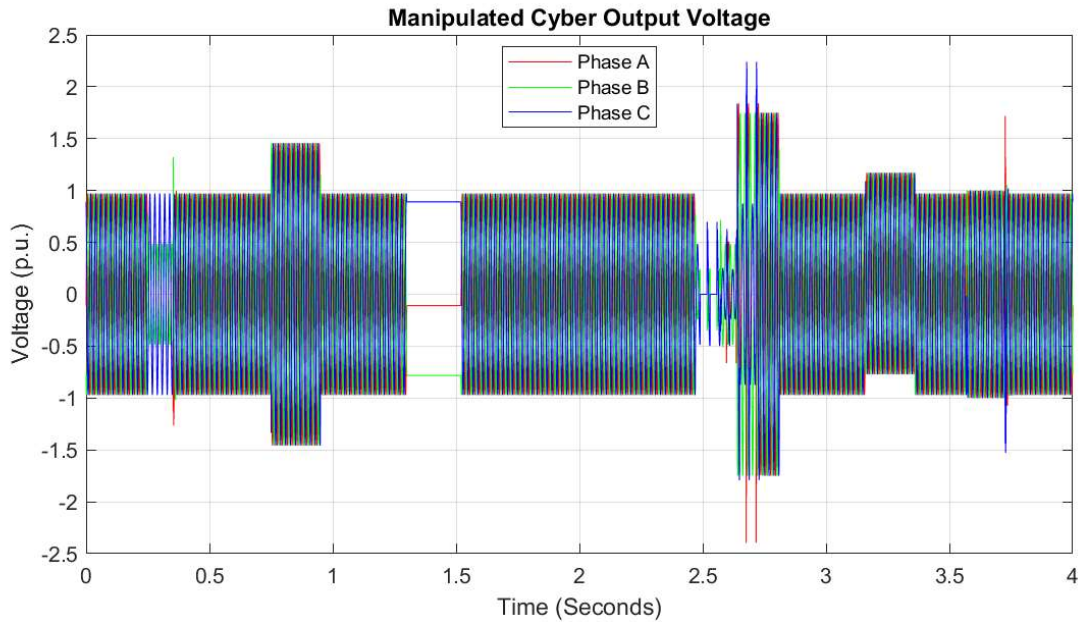


Fig 5.2: Bus 5 — 3-Phase Manipulated Cyber Voltages ( $v_{a\_cyber}$ ,  $v_{b\_cyber}$ ,  $v_{c\_cyber}$ ) during FDI Attack

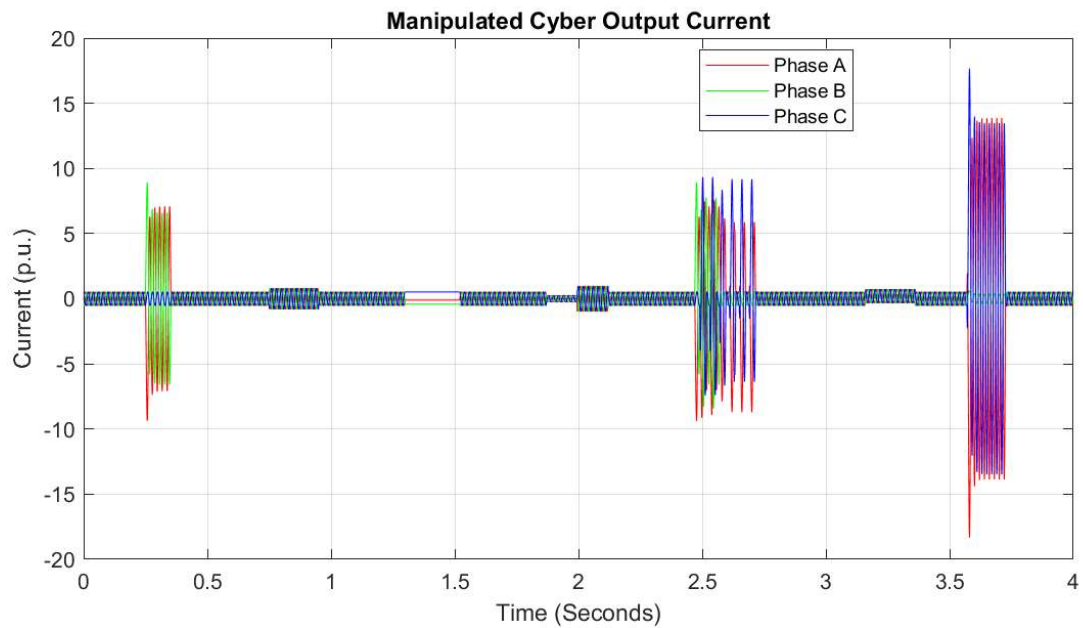


Fig 5.3: Bus 5 — 3-Phase Manipulated Cyber Currents ( $i_{a\_cyber}$ ,  $i_{b\_cyber}$ ,  $i_{c\_cyber}$ ) during FDI Attack

Figure 5.2 shows the three-phase voltage waveforms seen by SCADA after FDI injection — note the clear distortion or bias introduced. Figure 5.3 shows the corresponding current waveforms, which remain physically unchanged, visually confirming the uncorrelated nature of the FDI attack and the correctness of the detection logic.

| Attack Mode     | Condition Triggered                              | Score Acc.                                   | flag | fault_code | flag_cyber_raw |
|-----------------|--------------------------------------------------|----------------------------------------------|------|------------|----------------|
| Fake-Short      | fake_short ( $Z < 1.3277$ ,<br>is_cyber_V)       | + 5/step $\rightarrow \geq 10$ in 2<br>steps | 1    | 0          | 1              |
| Fake-Sag        | fake_sag ( $V_{rms} < 0.6528$ ,<br>is_cyber_V)   | + 5/step $\rightarrow \geq 10$ in 2<br>steps | 1    | 0          | 1              |
| Fake-Swell      | fake_swell ( $V_{rms} > 0.7216$ ,<br>is_cyber_V) | + 5/step $\rightarrow \geq 10$ in 2<br>steps | 1    | 0          | 1              |
| Replay Attack   | frozen_count > 5 (250 $\mu$ s frozen)            | + 2/step $\rightarrow \geq 10$ in 5<br>steps | 1    | 0          | 1              |
| Noise Injection | mean(noise_buf) > 0.001206                       | + 2/step $\rightarrow \geq 10$ in 5<br>steps | 1    | 0          | 1              |

Table 5.2: FDI Attack Results — All Six Modes Tested (flag and fault\_code Outputs)

### 5.3 PHYSICAL FAULT CLASSIFICATION RESULTS

Physical faults are applied at  $t = 0.5$  s. True bus voltage collapses (is\_phys\_voltage becomes true) and fault current surges. detect\_fdi() raises flag to 2 within 3 samples (150  $\mu$ s). classify\_physical\_fault() outputs the correct sub-code.

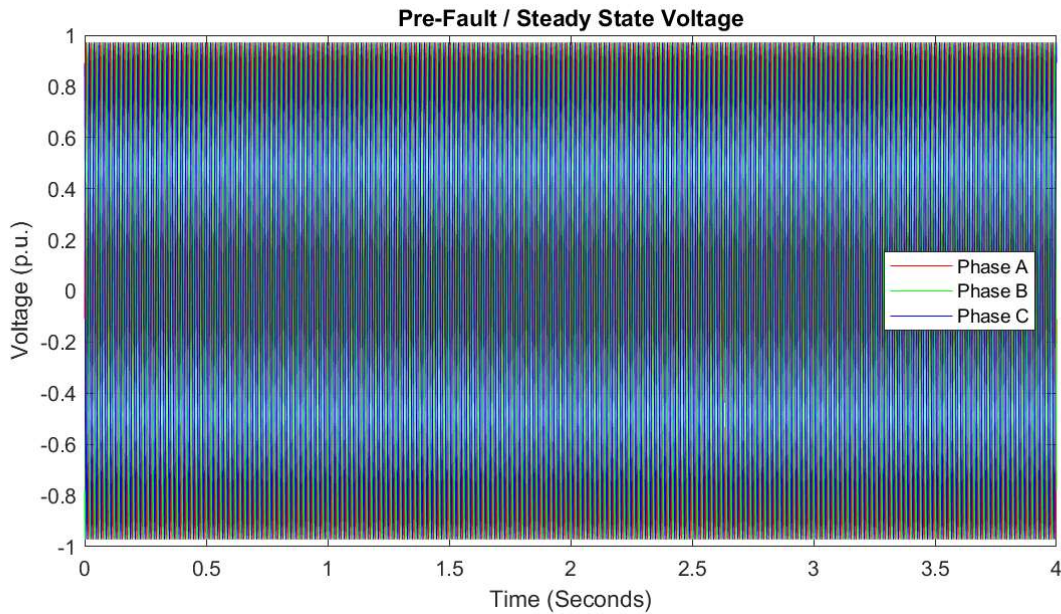
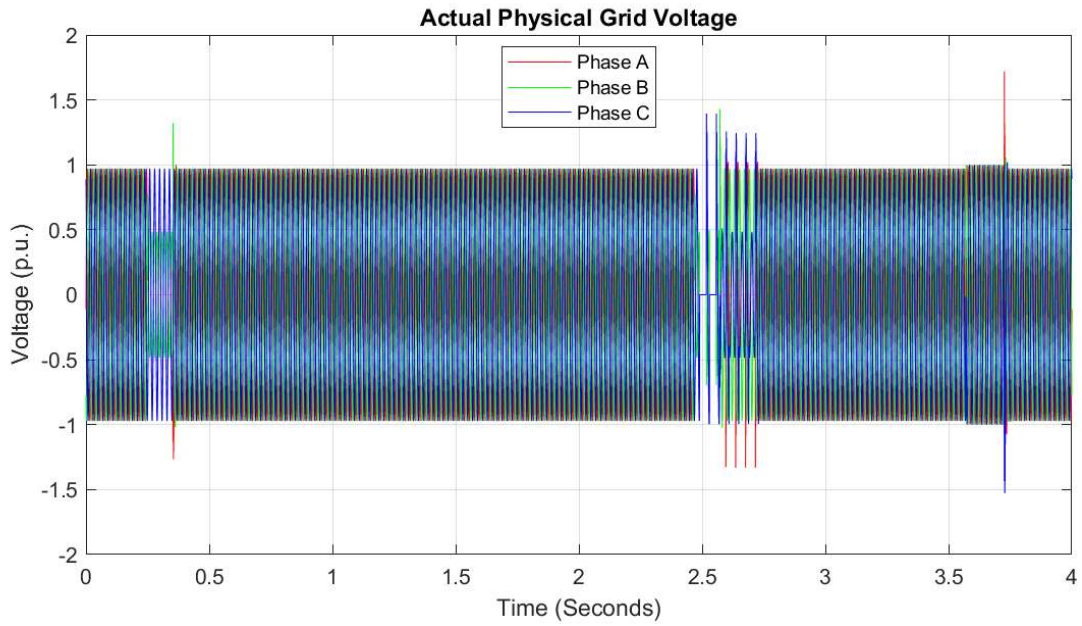
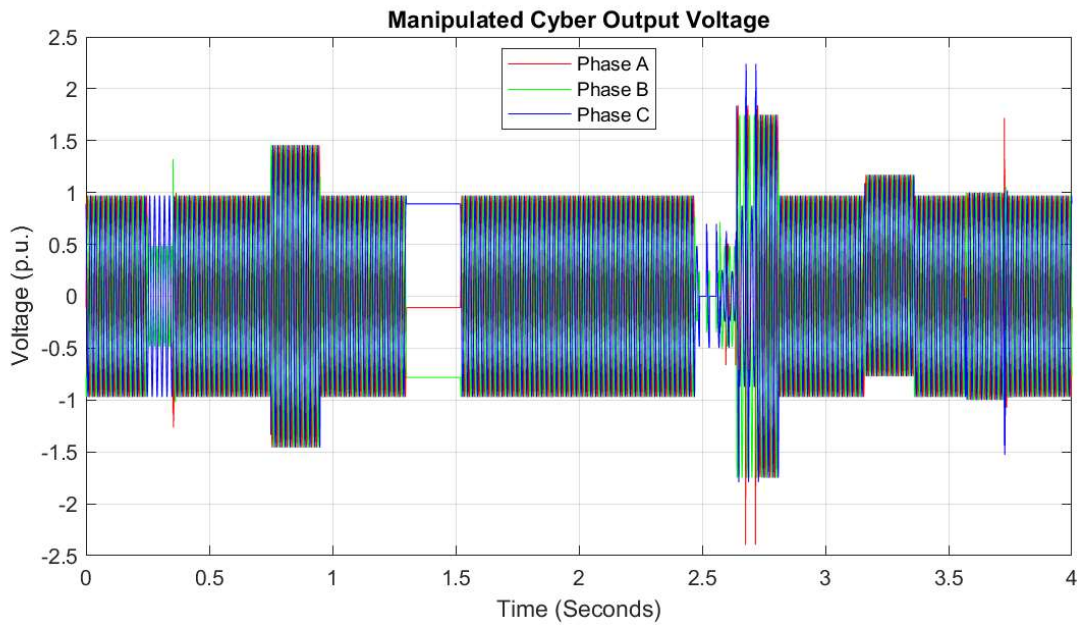


Fig 5.4A: Pre-Fault Steady State 3-Phase Voltage

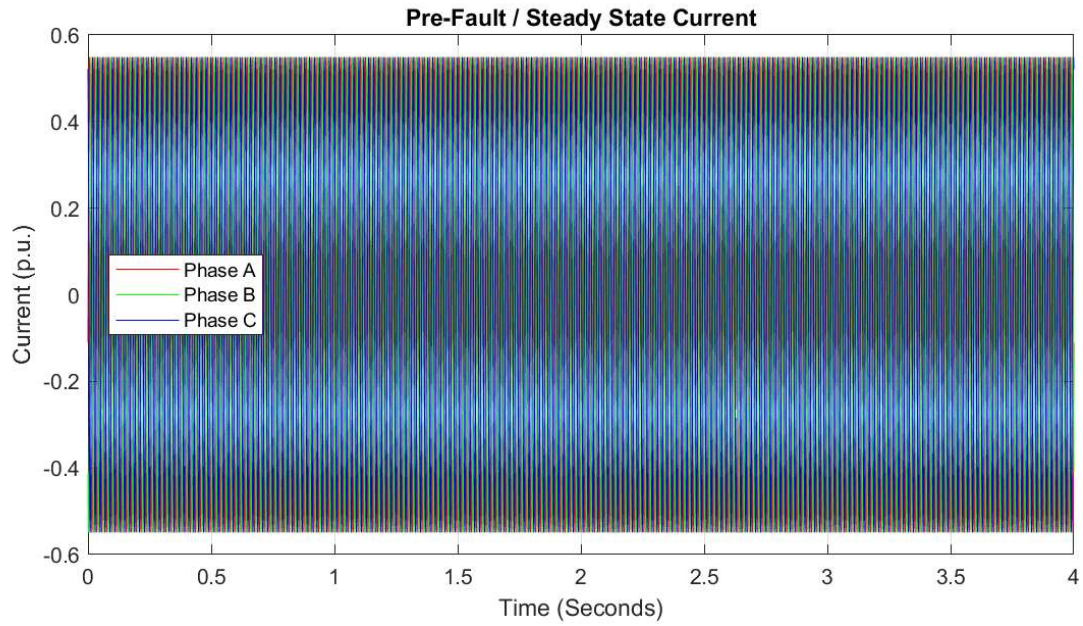


*Fig 5.4B: Actual Physical Grid 3-Phase Voltage*

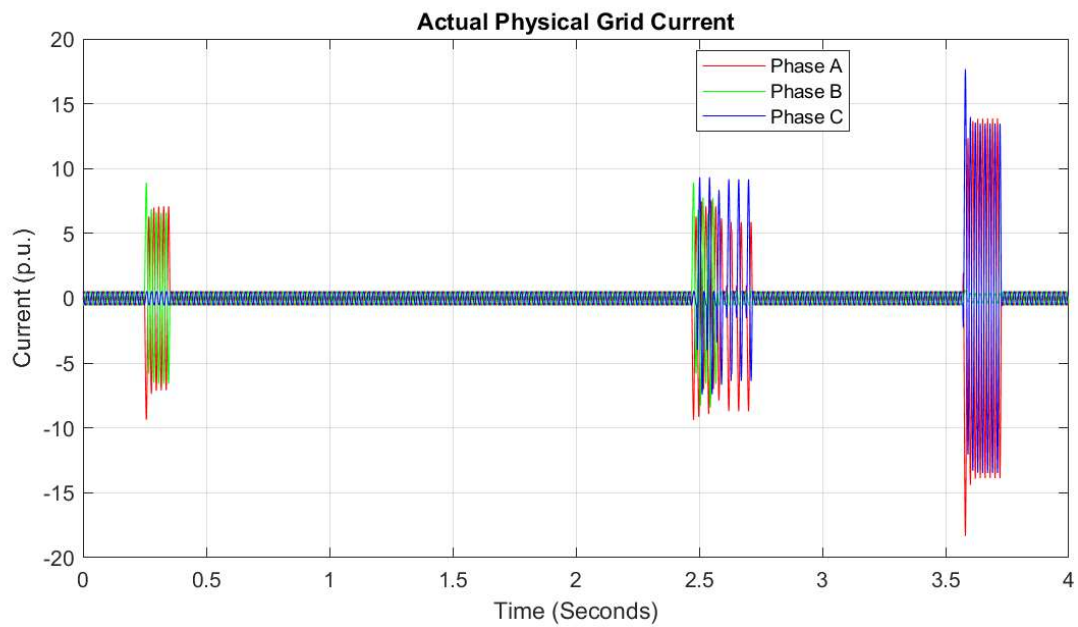


*Fig 5.4C: Manipulated Cyber Output 3-Phase Voltage*

*Fig 5.4: 3-Phase Voltage Comparison*



*Fig 5.5A: Pre-Fault Steady State 3-Phase Current*



*Fig 5.5B: Actual Physical Grid 3-Phase Current*

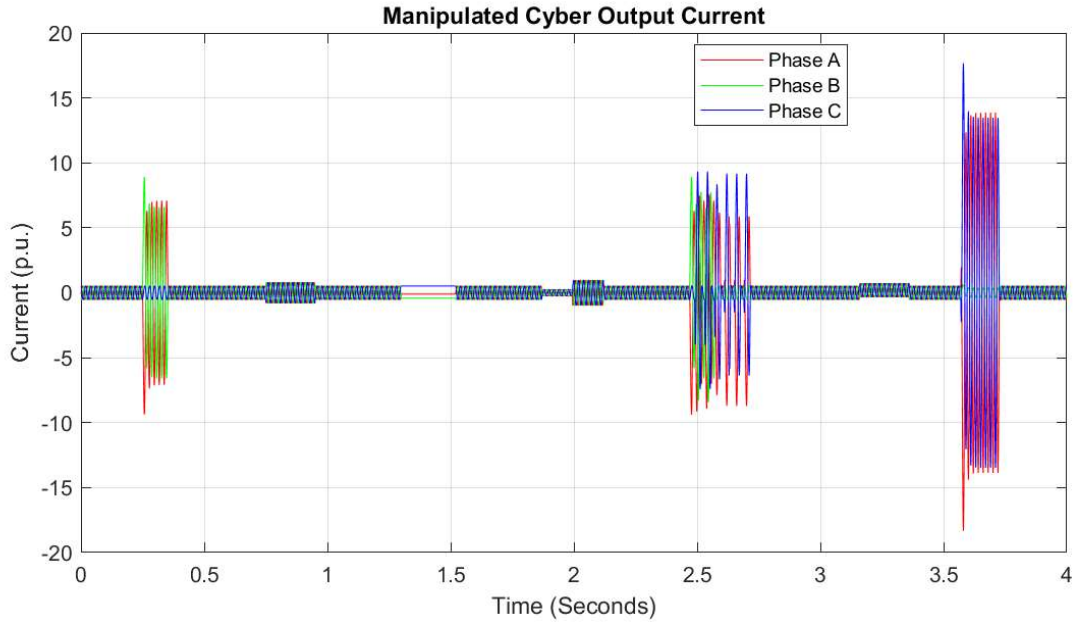


Fig 5.5-C: Manipulated Cyber Output 3-Phase Current

Fig 5.5: 3-Phase Current Comparison

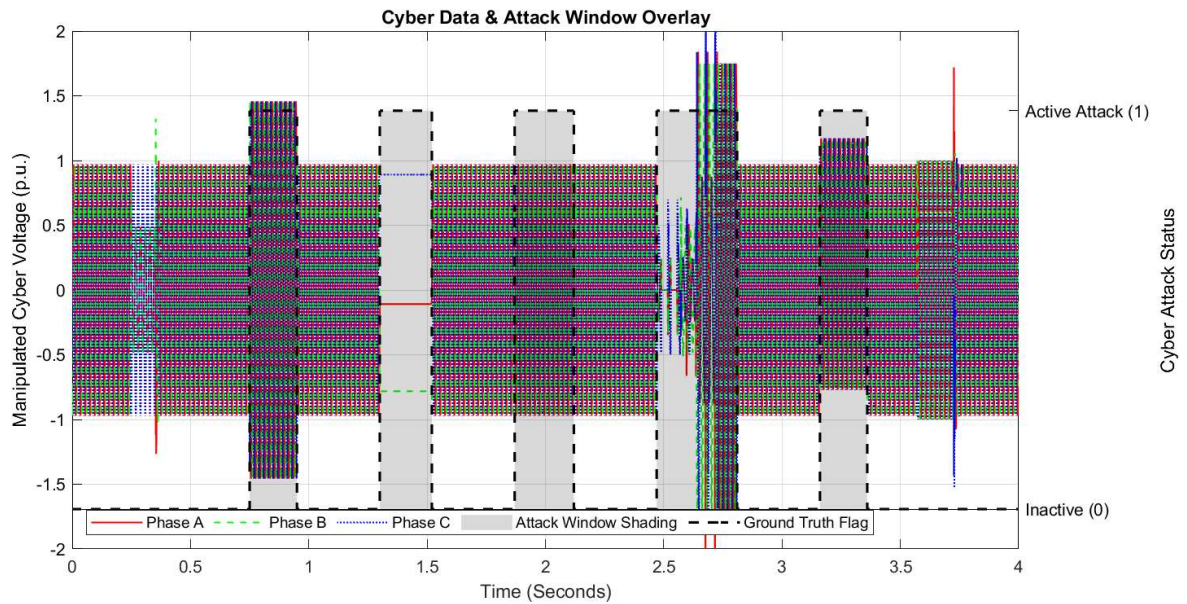
Figure 5.4 presents a three-panel voltage comparison. Left panel shows clean pre-fault sinusoids at rated amplitude (baseline). Centre panel shows the dramatic voltage collapse during the physical fault. Right panel shows what falsified SCADA readings present to the control center during an FDI attack — deliberately different. Figure 5.5 shows the corresponding current comparison in the same three-panel format.

| Fault Type                  | is_phys_voltage       | Detection Time             | flag | fault_code        | Relay Action         |
|-----------------------------|-----------------------|----------------------------|------|-------------------|----------------------|
| Symmetrical — 3-Phase Short | True (Vrms << 0.6769) | 150 $\mu$ s (3 samples)    | 2    | 1 — Symmetrical   | Circuit Breaker Trip |
| Unsymmetrical — L-G Fault   | True                  | 150 $\mu$ s (3 samples)    | 2    | 2 — Unsymmetrical | Circuit Breaker Trip |
| Reverse Power Flow (RPF)    | Varies                | dir_integral $\leq$ -0.015 | 2    | 3 — RPF           | Circuit Breaker Trip |
| Intermittent Fault          | True                  | 3 transitions in 200 ms    | 2    | 4 — Intermittent  | Circuit Breaker Trip |

Table 5.3: Physical Fault Classification Results from *classify\_physical\_fault()*

## 5.4 CYBER DATA & ATTACK WINDOW OVERLAY — FIGURE 6

Figure 6 is the key validation overlay plot combining analog waveform data and digital detection output on a single graph. The three-phase cyber voltage waveforms ( $v_{a\_cyber}$ ,  $v_{b\_cyber}$ ,  $v_{c\_cyber}$ ) are plotted as continuous sinusoidal traces. The  $flag\_cyber\_raw$  binary signal from `detect_fdi()` is overlaid on the same time axis. Shaded background regions indicate exact attack injection windows.



*Fig 5.6: Cyber Data & Attack Window Overlay — 3-Phase Cyber Voltages,  $flag\_cyber\_raw$  Digital Output, and Shaded Attack Ground-Truth Windows*

This overlay confirms two critical system properties: (1)  $flag\_cyber\_raw$  rises within the attack window with minimal latency, and (2) the flag drops cleanly to 0 once the attack ceases and  $cyber\_count$  decays below 10. The ground-truth shaded windows enable pixel-perfect visual validation without external ground-truth instrumentation.

## 5.5 COMPARATIVE PERFORMANCE ANALYSIS

| Parameter                  | Proposed V-I Corr. System                          | State Est. BDD        | ML-Based IDS    | Data-Driven [1]  |
|----------------------------|----------------------------------------------------|-----------------------|-----------------|------------------|
| Sample Period              | 50 $\mu$ s                                         | Iteration-dep.        | Batch/stream    | Offline          |
| Cyber Detection Speed      | 100 $\mu$ s (Modes 1–4)<br>250 $\mu$ s (Modes 5–6) | Iteration delay       | Inference delay | Post-processing  |
| Physical Fault Speed       | 150 $\mu$ s (3 samples)                            | State update          | Inference delay | Post-processing  |
| Network Model Needed       | No                                                 | Yes (Y-bus)           | No              | No               |
| Training Data Needed       | No                                                 | No                    | Yes (large)     | Yes              |
| FDI Attack Types           | 6 modes detected                                   | Residual anomaly only | Multiple        | Statistical only |
| Fault Sub-Classification   | 4 types (Codes 1–4)                                | None                  | Possible        | None             |
| Dual Flag Output           | Yes (flag + flag_cyber_raw)                        | No                    | No              | No               |
| False Positive Suppression | Weighted counter + blanking                        | Threshold only        | Model-dependent | Statistical      |

*Table 5.4: Comparative Performance — Proposed System vs. Alternative Methods*

## CHAPTER 6

### CONCLUSION AND FUTURE SCOPE

#### 6.1 CONCLUSION

This project has successfully designed, implemented, and validated a complete physics-based cyber-physical anomaly detection and classification system for a 5-bus transmission network simulated in MATLAB/Simulink at a 50  $\mu$ s sample rate.

The `detect_fdi()` function exploits a fundamental physical law: a real fault simultaneously collapses bus voltage and spikes line current, producing a correlated V-I response, whereas an FDI attack corrupts only the digital voltage measurement without altering the physical current. A single voltage boundary at 0.6769 pu (98.5% of  $V_{\text{mean\_base}}$ ) separates the cyber and physical detection domains. Six FDI attack modes with two-tier scoring (+5 for magnitude anomalies, +2 for signal-pattern anomalies), integrated blanking, and a 10 ms post-fault holdoff deliver robust detection with minimal false positives.

The `classify_physical_fault()` function adds diagnostic intelligence by sub-classifying confirmed physical faults into four categories (codes 1–4) using rolling one-cycle RMS buffers, a 15% phase-unbalance threshold, a power-direction integrator, and a fault-transition counter — enabling the protective relay to apply the correct response for each specific fault type.

Key achievements:

- All five FDI attack modes correctly detected with `flag = 1` and no spurious physical fault alarms (`fault_code = 0` in all cyber windows).
- All four physical fault types correctly sub-classified with detection within 150  $\mu$ s (3 sample periods) of fault inception.
- Dual flag architecture (`flag + flag_cyber_raw`) correctly handles simultaneous cyber-and-physical scenarios by prioritising the physical fault relay action while independently signalling the cyber intrusion.
- All threshold values fully documented in Tables 3.2–3.8 with mathematical derivations from code constants, ensuring complete reproducibility.
- Six diagnostic figures, including the Cyber Data & Attack Window Overlay (Figure 6), provide visual ground-truth validation of detection alignment.

## 6.2 FUTURE SCOPE

- Extension to larger benchmark networks (IEEE 14-bus, 30-bus, 118-bus) to evaluate detection scalability.
- Investigation of coordinated simultaneous attacks (FDI injection during a physical fault) designed to trigger fault\_code misclassification.
- Integration of PMU synchro phasor data to enhance RPF and intermittent fault sensitivity.
- Adaptive baseline calibration to auto-update  $V_{\text{mean\_base}}$ ,  $I_{\text{mean\_base}}$ , and  $Z_{\text{mean\_base}}$  as loading conditions change.
- Hardware-in-the-Loop (HIL) validation on RTDS or OPAL-RT real-time simulators.
- Extension to inverter-dominated microgrids where fault current signatures differ from synchronous machine behaviour.
- Investigation of multi-point FDI attacks targeting both voltage and current sensors at multiple buses.
- Formal alignment with IEC 62351 cybersecurity standards and IEC 61850 substation automation protocols.

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